



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

DIGITAL FORENSICS

System Forensics, Investigation, and Response

Lesson 11 Mobile Forensics

Learning Objective

- Summarize various types of digital forensics.

Key Concepts

- Mobile device concepts
- Evidence that can be obtained from a mobile device
- How to seize evidence from a mobile device

Cellular Device Concepts

Mobile switching center (MSC)

- The switching system for the cellular network

Base transceiver station (BTS)

- The part of the cellular network responsible for communications between the mobile phone and the network switching system

Home location register (HLR)

- A database used by the MSC that contains subscriber data and service information

Cellular Device Concepts (Cont.)

Subscriber identity module (SIM)

- A memory chip that stores the International Mobile Subscriber Identity (IMSI)

Electronic serial number (ESN)

- A unique identification number developed by the U.S. Federal Communications Commission (FCC) to identify cell phones

Personal unlocking code (PUK)

- A code used to reset a forgotten PIN

Network: Cellular



Wi-Fi

- Most cellular phones and other mobile devices can connect to Wi-Fi networks
- Free Wi-Fi hotspots in restaurants, coffee shops, hotels, homes, and many other locations

Operating Systems

iOS

- iPhone
- iPad
- iPod

Android

- Samsung Galaxy
- Many more

Windows 8

- Microsoft Mobile/
Nokia

Blackberry 10

- Blackberry

iOS

- Derived from OS X
- Interface based on touch and gestures
- In normal operations, iOS uses HFS+ file system
- Can use FAT32 when communicating with a PC

iOS (Cont.)

- Four layers:
 - **Core OS layer:** The heart of the operating system
 - **Core Services layer:** Where applications interact with the iOS
 - **Media layer:** Is responsible for music, video, and so on
 - **Cocoa Touch layer:** Responds to gestures

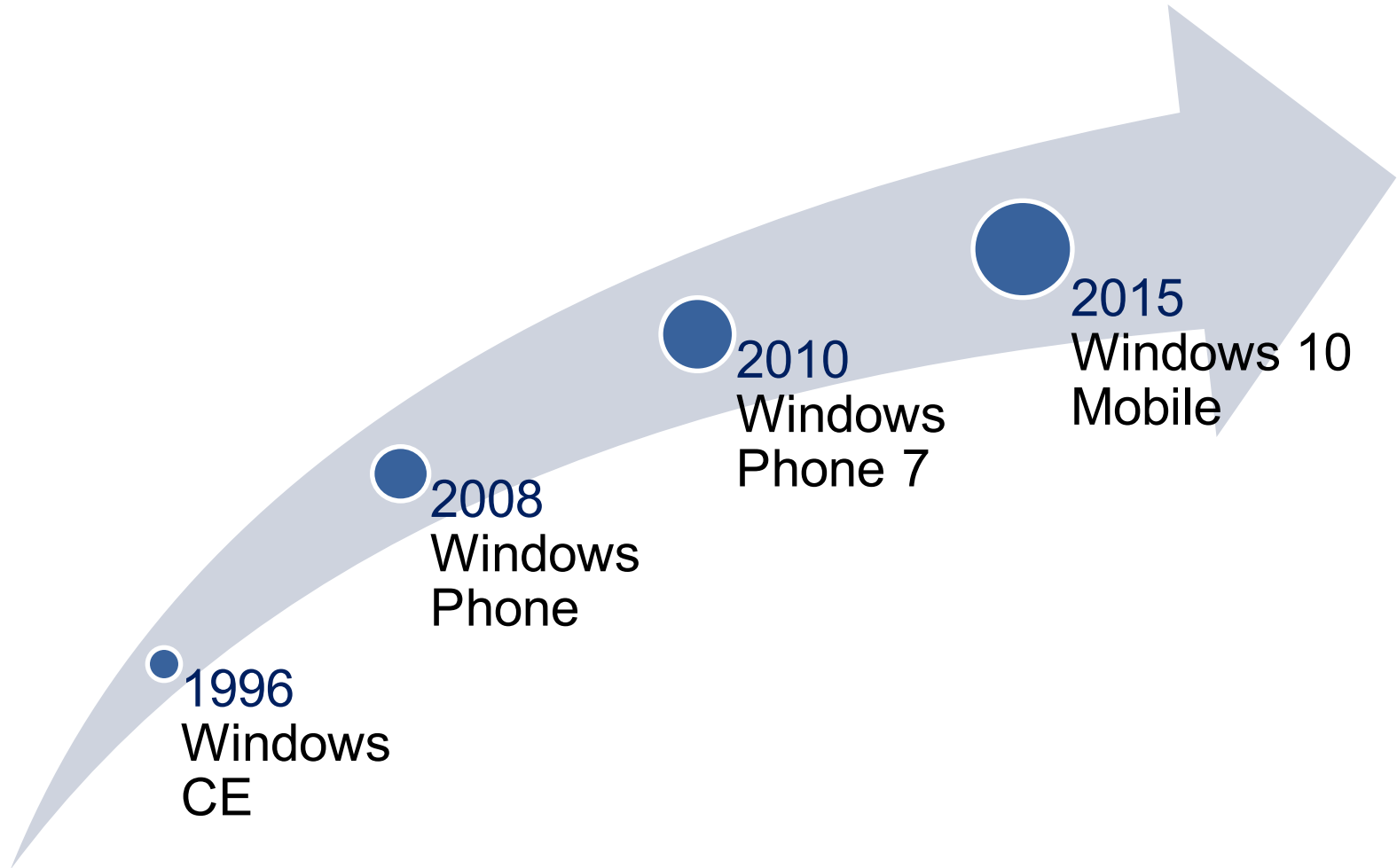
iOS (Cont.)

- Contains several elements in data partition:
 - Calendar entries
 - Contacts entries
 - Note entries
 - iPod_control directory (hidden)
 - iTunes configuration
 - iTunes music
- iPod_control\device\sysinfo folder contains model number and serial number

Android

- Linux-based operating system, completely open source
- First released in 2003
- Versions of Android named after sweets, such as Version 4.1–4.2 Jelly Bean and Version 7.0 Nougat
- Similarity across versions
 - Can perform similar forensic examinations on different versions

Windows



Blackberry 10

- Based on QRNX operating system
- Supports major features similar to other mobile phones
 - Drag and drop
 - Gestures

Evidence You Can Get from a Cell Phone

Call history

Emails, texts,
and/or other
messages

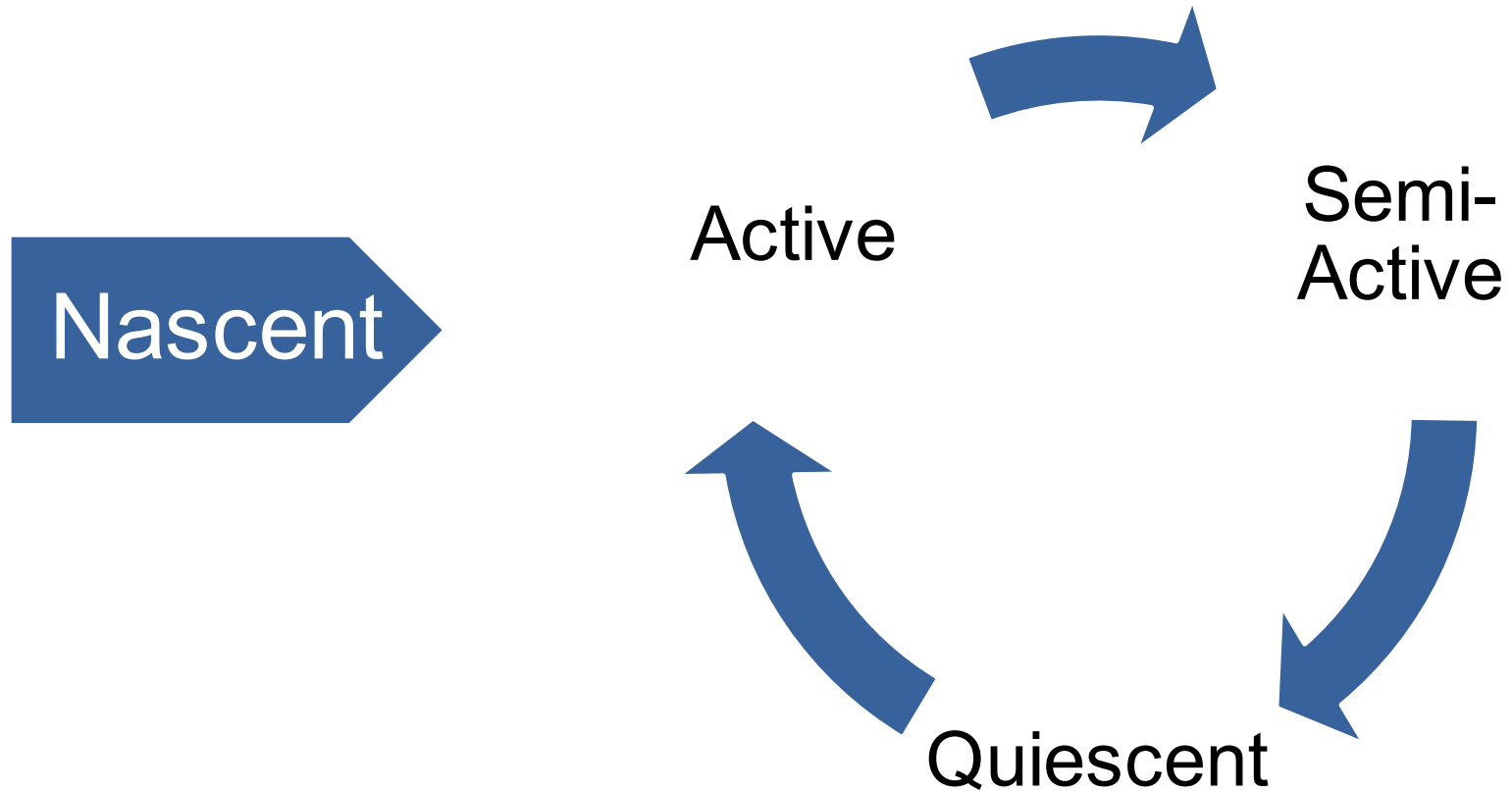
Photos and
video

Phone
information

GPS
information

Network
information

Mobile Device States



Rules for Seizing Evidence from a Mobile Device

- If you plug device into a computer, make sure device does not synchronize with the computer
- Touch evidence as little as possible
- Document what you do to the device
- Don't accidentally write data to the mobile device

Mobile Device Forensic Products

Oxygen
Forensics

Cellabrite

MobileEdit

Data
Doctor

Device
Seizure

Forensic
SIM Cloner

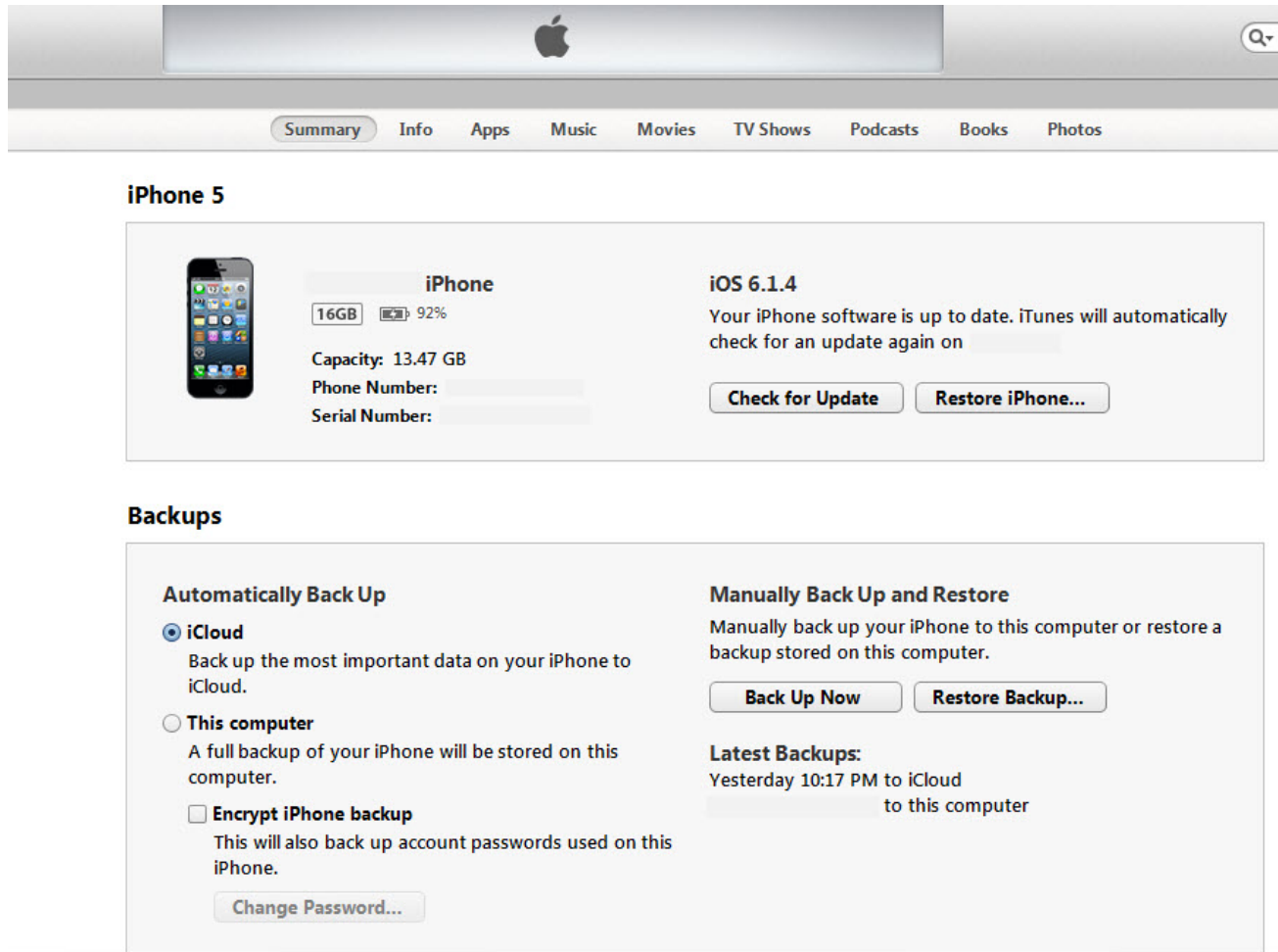
The iPhone: Seizing Evidence

- iPhone has four-digit pin
 - 10,000 possible combinations of the digits 0–9
- Can use automated process to break iPhone passcode, such as XRY
- Tools specifically for iOS devices:
 - Pwnage
 - Recover My iPod

The iPhone: Seizing Evidence (Cont.)

- If forensic workstation has iTunes:
 - Plug iPhone (or iPad/iPod) into the workstation
 - Use iTunes to extract information about the device

Apple iPhone iTunes Display



Screenshot republished with permission from Apple Inc.

Seizing Evidence from an iPhone

- Information from a device image:
 - Library_CallHistory_call_history.db
 - Contains entire call history
 - Library_Cookies_Cookies.plist
 - Contains cookies
 - Give you a history of the phone user's Internet activities

Seizing Evidence from an iPhone

- Information from a device image:
 - Library_Preferences_com.apple.mobileipod.plist
 - Library_Preferences_com.apple.mobileemail.plist
 - Gives you information about email sent and received from the phone
 - Library_Preferences_com.apple.mobilevpn.plist
 - Indicates if user used device to communicate over a VPN

Seizing Evidence from an iPhone

- Deleted files
 - When a file is deleted on iPhone/iPad/iPod, moved to.Trashes\501 folder
 - Data exists until overwritten

Seizing Evidence from a BlackBerry

- Download and install BlackBerry Desktop Manager
- Steps to create complete backup image:
 1. Open BlackBerry's Desktop Manager. Click Options then Connection Settings.
 2. If the Desktop Manager hasn't already done so, select USB-PIN: Device # for connection type. Click OK.

Seizing Evidence from a Blackberry (Cont.)

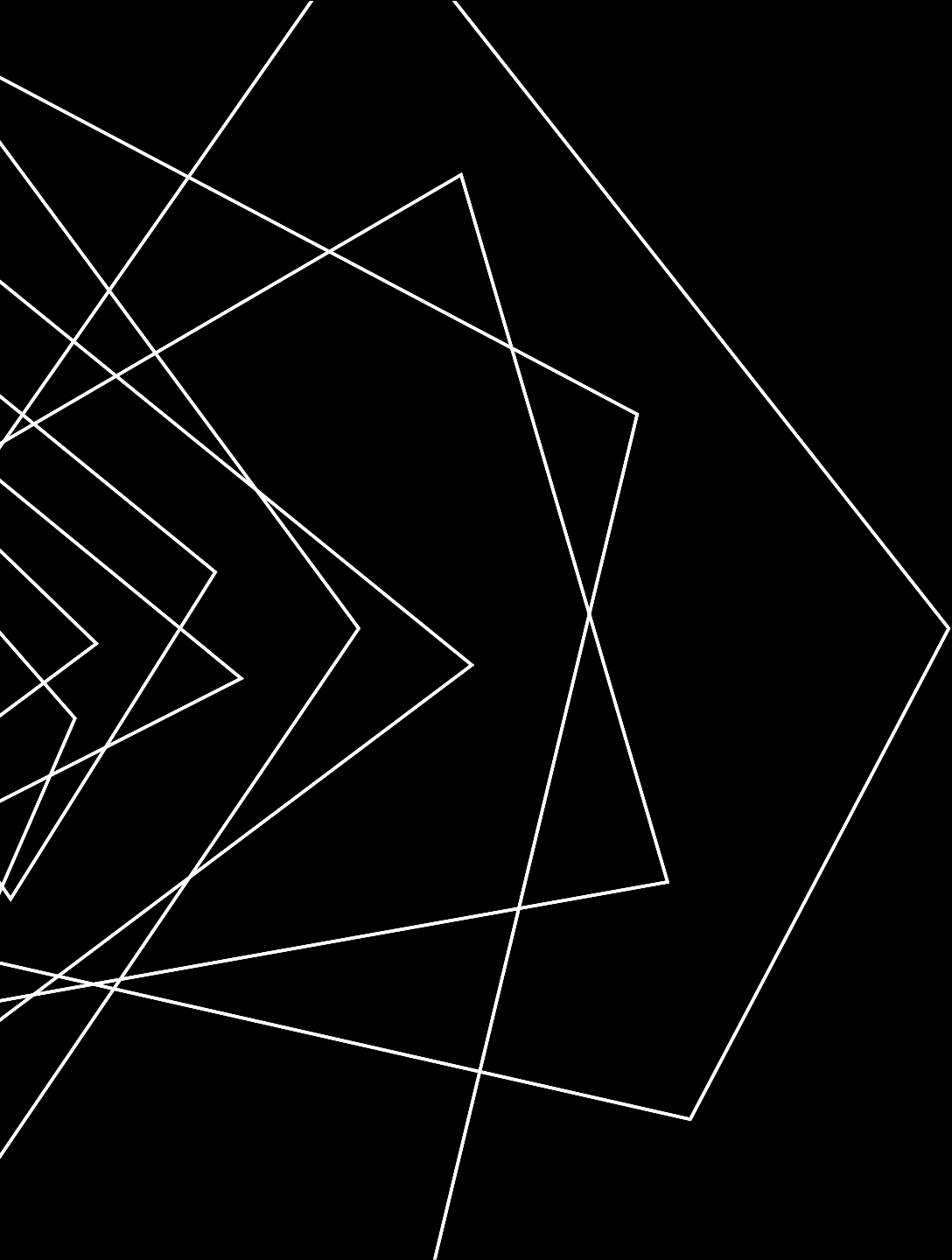
3. Select Backup and Restore.
4. Click the Back Up button for a full backup of the device or use the Advanced section for specific data.
5. Select your destination (such as workstation) and save the .ipd file.
6. Examine data and perform a forensic analysis.

JTAG

- Joint Test Action Group (JTAG)
- An Institute of Electrical and Electronics Engineers (IEEE) standard for testing chips
- Test access points (TAPs) used to directly access the chip and extract data
- Forensic examiner takes back off of phone, and then connects wires by soldering or by using some other means to the TAPs of the phone's memory chip
- Wires also connected to a JTAG device that uses software to extract the data directly from the memory chip

Summary

- Mobile device concepts
- Evidence that can be obtained from a mobile device
- How to seize evidence from a mobile device



THANK YOU

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